

```

20 21 22 23 24 25
2    3
3    5 2
4    0 3 5
5   29 26 24 29
6    5 2 0 5 24
7    6 3 1 6 23 1
8    9 6 4 9 20 4 3
9   11 8 6 11 18 6 5 2
10  13 10 8 13 16 8 7 4 2
11  9 6 4 9 20 4 3 0 2 4
12  13 10 8 13 16 8 7 4 2 0 4
13  6 3 1 6 23 1 0 3 5 7 3 7
14  10 13 15 10 39 15 16 19 21 23 19 23 16
15  1 2 4 1 28 4 5 8 10 12 8 12 5 11
16  10 7 5 10 19 5 4 1 1 3 1 3 4 20 9
17  26 23 21 26 3 21 20 17 15 13 17 13 20 36 25 16
18  7 4 2 7 22 2 1 2 4 6 2 6 1 17 6 3 19
19  5 8 10 5 34 10 11 14 16 18 14 18 11 5 6 15 31 12
20  1 2 4 1 28 4 5 8 10 12 8 12 5 11 0 9 25 6 6
21  8 5 3 8 21 3 2 1 3 5 1 5 2 18 7 2 18 1
13 7
22 12 9 7 12 17 7 6 3 1 1 3 1 6 22 11 2 14 5 17
11 4
23 14 11 9 14 15 9 8 5 3 1 5 1 8 24 13 4 12 7 19
13 6 2
    26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44
45 46 47 48 49 50
...

```

The binary distance measurement output for the same bank age data is as follows:

```

> dist(bank$age, method="binary", diag=FALSE, upper=FALSE,
p=2)
 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23 24 25 26 27 28
2    0
3    0 0
4    0 0 0
5    0 0 0 0
6    0 0 0 0 0
7    0 0 0 0 0 0
8    0 0 0 0 0 0 0
9    0 0 0 0 0 0 0 0
10   0 0 0 0 0 0 0 0 0
11   0 0 0 0 0 0 0 0 0 0
12   0 0 0 0 0 0 0 0 0 0 0
13   0 0 0 0 0 0 0 0 0 0 0 0
14   0 0 0 0 0 0 0 0 0 0 0 0 0
15   0 0 0 0 0 0 0 0 0 0 0 0 0 0
16   0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

```

```

17  0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
18  0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
19  0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
20  0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
21  0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
22  0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
23  0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
    29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47
48 49 50 51 52 53

```

Quantile

The quantiles can be generated for a numeric vector 'x' along with their given probabilities using the *quantile* function. The 'NA' and 'NaN' values are not allowed in the vector, unless 'na.rm' is set to TRUE. The probability of 0 is for the lowest observation and 1 for the highest. Its syntax is as follows:

```
quantile(x, ...)
```

The quantile function accepts the following arguments:

Argument	Description
<i>x</i>	A numeric vector
<i>probs</i>	A vector of probabilities between 0 and 1
<i>na.rm</i>	NA and NaN values are ignored if TRUE
<i>names</i>	If TRUE, the result has the names attribute
<i>type</i>	An integer to select any of the nine quantile algorithms
<i>digits</i>	The decimal precision to use
...	Additional arguments passed to or from other methods

The *rnorm* function can be used to generate random values for the normal distribution. It can take 'n' number of observations, a vector of 'means', and a vector 'sd' of standard deviations as its arguments. We can obtain the quantiles for the values produced by the 'rnorm' function. For example:

```

> quantile(x <- rnorm(100))
      0%      25%      50%      75%     100%
-1.978171612 -0.746829079 -0.009440368  0.698271134
1.897942805

```

The quantiles for bank age data with the following probabilities can be generated as follows: